# SRW3 Phase 1J — Security Closure Report

**Reachability Safety Closure & Execution-Evidence Completeness**

Branch `phase1j-security-closure` · additive tree `phase1j/` · frozen base `8a520fd8a3d4651b0884a59e8db8aec1d3eb7670` (`phase1i-r3-complete`)

## 1. Objective and outcome

Phase 1J advances SRW3 from the verified configuration commitment (1I-R3) to an explicit **reachable-committed-state safety theorem** over a new isolated transition system `M_J`. The theorem target — `J-REACHABLE-COMMITTED-SAFE` (every committed prefix of every reachable finite execution satisfies the independently defined safety predicate `Safe_P`) — is decomposed per the handoff §3 into base, execution/evidence, effect closure, catalog integrity, interaction closure, preservation, gate/commit bridge, and induction obligations.

**Outcome (Level III, conditional): CLOSED FOR THE DECLARED FINITE FRAGMENT, conditional on the listed anchor/source assumptions — with four retained not-mechanized items whose concrete substitutes are delivered.** The K ladder is **38 PROVED / 4 retained** over a 42-claim inventory; the Python adversarial suite is **48/48 PASS** over all 24 normative attack classes plus honest pipelines, determinism, and golden vectors; the LLVM suite (real Ethereum Keccak-256 via the Phase-1C shim) is **14 executed demos + 5 kparse ill-formedness rejections**; the cross-layer conformance is **22/22 byte-identical** fields; all frozen regression suites reproduce exactly and the frozen trees are byte-identical to the R3 base.

The four retained items are the honest Haskell-boundary outcomes, each with its failed log retained and its concrete substitute delivered (`claim-classification.csv`): `J-EVIDENCE-STATE-CORRESPONDENCE` and the two pipeline claims (the keccak hook has no evaluator on the hs backend — the documented R3 `PIPELINE-HONEST` class; demonstrated on LLVM T1/T2/T3/T6 and checked in Python), and `J-REACHABLE-COMMITTED-SAFE` (the kore stuck-configuration/circularity blocker — the frozen `PI7-ALL` class; the closure evidence is the proved component lemmas, the Python exhaustive sequence enumeration — 3474 committed prefixes, 386 unsafe derived states caught by `Safe_P`, zero committed unsafe — and the LLVM two-block pipeline).

## 2. What was built

`phase1j/semantics/srw3closure-j.k` — the ISOLATED J machine (module `SRW3CLOSURE-J`, 2069 lines), importing ONLY the frozen Phase 1G gate:

- **The concrete policy catalog** (handoff §5.1): registered applications, resources (the flat 64-bit 1E key space, 8-byte canonical encoding), n-ary hyperedges, and a DECLARED finite typed invariant fragment (local bounds; sum over a key set; equality over a key set) organized as local / hyperedge / global classes. Canonical encoding with a new domain (`"SRW3/PolicyCat/J"||0x00`, versioned, length-delimited, deterministic order) and three content-class sub-roots (0xB1/0xB2/0xB3) that the pin binds to the R3-rooted configuration fields 3/4/5 — **catalog CONTENTS are authenticated by the same R3 anchor** (the digest-without-contents and contents-without-digest refusals both fail closed).
- **The closed execution fragment** `EVM_F` (Option A, handoff §4.1): a four-op interpreter (Read/Write/Call/Return — the frozen 1F vocabulary) that DERIVES the ordered event trace, the effective write-delta, and the post-state from the current abstract state. Excluded effect classes and out-of-range operands produce EXPLICIT refusals (`jExecUnsupported`/ `jExecBounds`) — never skipped, never silent no-ops. The trace is emitted by the semantics (the only trace producers are the interpreter rules); the footprint is the first-occurrence projection of the trace.
- **The J state root** (`StateRootJ`, tag 0xB4 over the frozen ascending map canon) and the **evidence identity** (`EvidenceIDJ`, new domain `"SRW3/Evidence/J"`, binding config root, catalog root, chain/fork, parent/post state roots, payload digest, execution-source identity, trace digest, footprint digest, policy-context digest).
- **Two receipt kinds**: the exec receipt (minted ONLY by `pExecJ`, all fields machine-derived) and the gate receipt (minted ONLY by `pGateEvalJ`, verdict COMPUTED by the frozen 21-conjunct gate over the machine-constructed context, embedding the exec receipt it was derived from). Both consumed exactly once by the commit.
- **The bounded hypergraph closure** (handoff §5.2): the least closure of the touched keys plus every incident hyperedge plus the additional nodes needed to evaluate each incident invariant, under an explicit fuel bound; exhaustion refuses (never "no more interactions"); unregistered touched keys refuse.
- `Safe_P` **computed, independent of the gate** (handoff §3/§6): `SafeJ` evaluates every registered invariant on a state — used at the pin (the genesis must be safe: `J-INIT-SAFE` enforced at init) and at the commit boundary on the DERIVED post-state.

- **One chain-extending rule** (`pCommitJ`, handoff §7) with the nine requirement classes in the R3 guard order (decidable conjuncts first, the computed policy result, the hash-class conjuncts last); rejections leave every cell unchanged.

`phase1j/semantics/srw3closure-j-demo.k` — the fixture suite: the 3-app countermodel catalog (oracle/lending/vault with the 3-app solvency hyperedge `he1={k1,k2,k3}` and the pairwise edge `he2={k2,k4}`), the honest evidence chain BUILT FROM the derived artifacts (the correspondence is structural), the mutation/malformation fixtures, and the cross-layer byte certificate (`xCertJ`).

`phase1j/proofs/` — the 42-claim ladder (`j_proofs.k`), the runner (`run_j_proofs.sh`, resumable, per-claim exit codes + retained logs), the all-literal fixtures (`j_fixtures.k`), and the LLVM build script (`j_build_llvm.sh`).

`phase1j/python/` — the executable mirror (`j_model.py`: catalog canon + roots, fragment interpreter, footprint, state roots, evidence identity, invariant evaluation, bounded closure, the full machine driving the REAL frozen gate via the unchanged Phase-1G model) and the adversarial runner (`run_j_attacks.py`: J-A01..J-A24 + honest pipelines + determinism + golden vectors + the reachability sequence enumeration).

`phase1j/tests/`, `phase1j/vectors/`, `phase1j/transcripts/{k,llvm,attacks,regression}/`, `phase1j/audit/`, `phase1j/report/`, `phase1j/provenance/` — per the handoff §11 layout.

## 3. The theorem, honestly stated

For the declared fragment and the fixture catalog, the following are MECHANIZED (hs backend, logs retained):

- **Base.** `J-INIT-SAFE-BASE`: the registered invariants hold on the genesis state; an unsafe genesis is REFUSED at the pin (`J-INIT-REFUSES-UNSAFE-GENESIS`).
- **Origin/correspondence structure.** The trace is emitted by the semantics (`J-FRAG-TRACE-WRITE/ORDER/READ-AFTER-WRITE`); the footprint is the derived projection (`J-FOOTPRINT-PROJECTION`); the candidate identity fields equal the derived values (the mint's binding conjuncts — demonstrated concretely on LLVM where the mints fire, and in Python).
- **Effect closure.** Every excluded class refuses (`J-FRAG-UNSUPPORTED/-MIDPROGRAM/-BOUNDS`); nothing outside the fragment can silently affect the state.

- **Catalog integrity.** The canonical preimage is pinned byte-for-byte (`J-CATALOG-CANON-FIXTURE`, the golden vector); each content class binds (`J-CATALOG-BINDS-APPS/RESOURCES/GRAPH/POLICY`); well-formedness is total and fail-closed (`J-CATALOG-WF-*`); the missing 3-app hyperedge changes the preimage and refuses the pin (`J-CATALOG-BINDS-GRAPH`, `J-INIT-REFUSES-GRAPH-MISMATCH`).
- **Interaction closure.** `J-CLOSURE-FIXTURE` (the fixed point pulls in the pairwise edge via the 3-app edge) and `J-CLOSURE-EXHAUSTED` (bounded exhaustion refuses); unknown touched keys refuse (`J-FOOTPRINT-REGISTERED-UNKNOWN`).
- **Safety decomposition.** `J-SAFE-HONEST`; the 3-app countermodel `J-SAFE-UNSAFE-COMPOSITION` (locals pass, the pairwise invariant passes, the 3-app hyperedge invariant FAILS, `SafeJ` false); the global class `J-SAFE-GLOBAL-FAIL`; `J-FRAME-PRESERVATION`.
- **Gate/commit bridge.** `J-GATELINK-VALID` (the derived-decision link), `J-DECISION-BOUND` (only "valid-g" is protocol-valid), `J-FORGED-VERDICT-REJECT` (a non-valid-g computed verdict cannot commit), `J-COMMIT-NO-RECEIPT-REJECTS`; the anchor discipline (`J-INIT-PINS-AUTHORIZED`, `J-INIT-ROOT-MISMATCH-REJECTS`).

**Not mechanized (retained, with substitutes):** the record/digest-level correspondence and the two full-pipeline one-step claims (hs keccak boundary — demonstrated on LLVM T1/T2/T3/T6, checked in Python), and the universal reachability induction (the kore circularity class — the closure evidence is the component lemmas above plus the Python exhaustive sequence enumeration over the closed fragment: 3474 committed prefixes checked, 386 unsafe derived states caught by `Safe_P`, zero committed unsafe, plus the LLVM two-block honest pipeline). Per the handoff §10, a failed universal theorem accurately recorded is not a project failure; relabeling it would be.

## 4. The cross-layer evidence

- **22/22 byte-identical** conformance fields between the K LLVM certificate (real Keccak-256) and the Python mirror: catalog canon bytes and length, catalog root, three sub-roots, config root, three state roots, trace digest, footprint digest + canon, payload digest, policy-context digest, evidence identity, block commitment, certificate id — plus the computed verdicts (verdict0/verdict1 "valid-g") and the SafeJ evaluations, with the disclosed verdict-surface normalization (the R2/R3 mirror artifact: the Python gate emits layer-coded verdicts; `verdict_k` normalizes; only "valid-g" maps to the valid decision on both

sides).

- The machine's honest pipeline executes END-TO-END on LLVM with REAL keccak: init (anchor check) → exec (derived receipt) → gate (computed verdict) → commit (state moves to the derived post-state) — twice (T2), with the unsafe third block REJECTED (T6: state, head, and counter unchanged; the receipts stay unconsumed).
- The Python mirror drives the REAL frozen gate (`verify_lineage_g` imported unchanged from the Phase-1G model) — including the demonstrative fact that the gate ACCEPTS the unsafe record (`J-A13`: "valid-g"): the authority layer and the policy layer are DISTINCT, and the commit boundary composes them fail-closed.

## 5. Assumptions, exclusions, and costs

(See `audit/SECURITY-CLOSURE-AUDIT.md` §E for the full register.)

- **ASSUMED**: the authenticated anchor; Keccak-256 collision resistance; the frozen gate's soundness as used; the invariant language IS the intended policy (the typed fragment); the declared execution source's integrity; the 1F/1G input-side authenticity boundary.
- **EXCLUDED and fail-closed**: LOGn, CREATE/CREATE2, CALLCODE, DELEGATECALL, STATICCALL, SELFDESTRUCT, gas accounting, memory operations, precompile/system effects — each is an explicit refusal (`jUnsupported`), never a silent no-op. Option B (full-EVM evidence) remains OPEN and is not claimed.
- **Liveness cost**: fail-closed availability — unavailable evidence, catalog, or closure data means the commit never fires (a non-committing defer); unavailability is never interpreted as acceptance.
- **No consensus-integration claim** (Level III model; the 1H Geth adapter was a shadow boundary and remains one). The disposition is: **closed for the declared finite effect/policy fragment, conditional on the listed anchor/source assumptions** — NOT "end-to-end security closed".

## 6. Regression and provenance

- Frozen trees byte-identical to `8a520fd` (git diff over everything except `phase1j/` is empty).
- Frozen suites reproduce exactly: 1E baseline OK; 1F 16/16; 1G 25/25; R1 48/48; R2 87/87; R3 181/181.
- 1J Python: 48/48 adversarial checks; 27/27 unittests.
- `provenance/MANIFEST.txt` + `provenance/provenance-hashes.txt` record the

phase1j tree (107 files hashed at packaging time).

## 7. Artifact index

```
phase1j/
semantics/srw3closure-j.k the isolated J machine
semantics/srw3closure-j-demo.k fixtures + the cross-layer certificate
proofs/j_fixtures.k proof-definition fixtures (inert)
proofs/j_proofs.k the 42-claim ladder
proofs/run_j_proofs.sh the resumable runner (score 1:1)
proofs/j_build_llvm.sh the LLVM + krypto-shim build
python/j_model.py the executable mirror (real keccak)
python/run_j_attacks.py J-A01..A24 + pipelines + vectors
python/pack_cfg_shim.py the 13-field config builder shim
tests/test_security_closure.py 27 unittests
vectors/j_golden.json the cross-layer conformance vectors
transcripts/k/ the K ladder: logs + claims-status.csv
+ claim-classification.csv + score.txt
transcripts/llvm/ T0-T15 (14 executed + 5 kparse rejects)
transcripts/attacks/run_j_attacks.txt
transcripts/regression/regression.txt
audit/SECURITY-CLOSURE-AUDIT.md
report/SRW3-Phase1J-Security-Closure-Report.{md,pdf}
provenance/MANIFEST.txt + provenance-hashes.txt
```